

Team Contribution Report

Project Information

Project Title: Airport Duty-Free Shop Management System

Team Leader: Alexandra

Team Members:

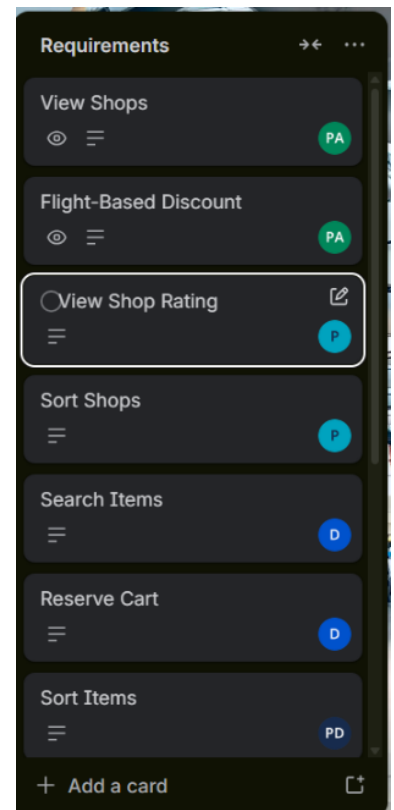
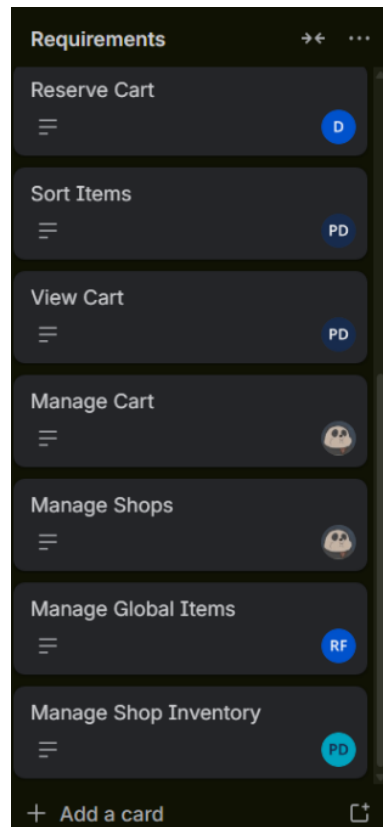
- Alex
- Andrada
- David
- Denisa
- Felician
- Maya

1. Task Distribution

Team Member Responsibilities

Alexandra	View Shops + Flight-Based Discount
Alex	View Shop Ratings + Sort Shops
Andrada	Manage Cart + Manage Shops
David	Search Items + Reserve Cart
Denisa	Manage Shop Inventory
Felician	Manage Global Items
Maya	Sort Items + View Cart

Screenshot of task assignment board:

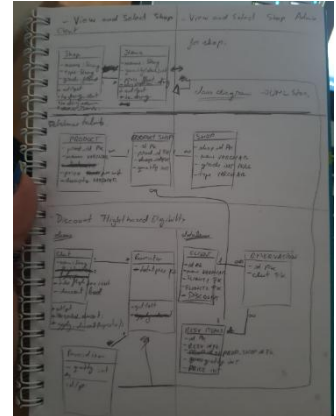


2. Individual Contributions

2.1 Alexandra - View Shops + Flight-Based Discount

Tasks Completed

- Create database tables for given requirements
- Created classes for given requirements
- Create tasks for given requirements (Trello)



2.2 Alex - View Shop Ratings + Sort Shops

Tasks Completed

- Create database tables for given requirements (partial)

Tasks Not Completed

- Created classes for given requirements
- Create tasks for given requirements (Trello)

```
create table Shops(
shop_id INT PRIMARY KEY IDENTIFY (1,1),
shop_name VARCHAR(100) NOT NULL,
shop_type VARCHAR(10) NOT NULL,
manager_id INT FOREIGN KEY REFERENCES Managers(manager_id),
shop_score INT NOT NULL
);
```

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la shop score trebuie sa ma mai gandesc putin 21:42

2.3 Andrada - Manage Cart + Manage Shops

Tasks Completed

- Create database tables for given requirements
- Created classes for given requirements
- Create tasks for given requirements (Trello)

Andrada

Items Table
This represents the "Shop" inventory. It must store the master data for validation.
item_id: Unique Identifier (Primary Key).
item_name: The name displayed in the cart.
unit_price: The current price per single unit.
available_stock: The total quantity currently in the shop. This is used to verify that a client doesn't add more than what is available.

Carts Table
This table acts as a "container" that links a specific client to their current shopping session.
cart_id: Unique Identifier (Primary Key).
user_id: Links the cart to a specific Client (Foreign Key).
status: (Optional) To distinguish between an "Active" cart and a "Checked Out" cart.

CartItems Table
This it tracks the specific interaction between the client and the items
cart_item_id: Unique Identifier.
cart_id: Links this entry to the specific user's cart.
item_id: Links to the item being added.
quantity_selected: The number of items the client wants to buy.

Andrada

User Class
Represents the client interacting with the system.

Attributes:
userid: int
username: String
email: String

Methods:
getCart(): Cart
checkout(): boolean

Item Class
Represents the products available in the shop inventory.

Attributes:
itemid: int
itemName: String
unitPrice: double
availableStock: int

Methods:
hasSufficientStock(requestedQuantity: int): boolean
reserveStock(quantity: int): void (Called only during checkout)
releaseStock(quantity: int): void (Called if a reservation is cancelled)

Cart Class
The active shopping session linked to a specific user.

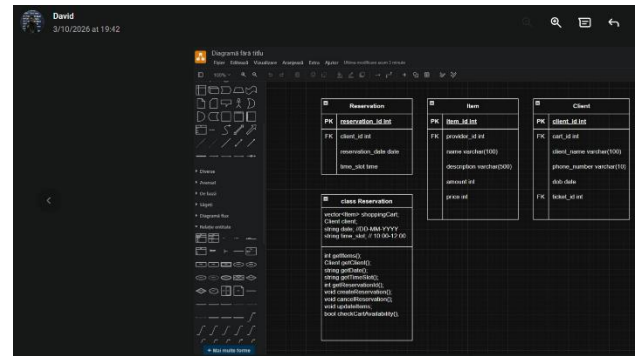
Attributes:
cartId: int
userId: int
Items: List<CartItem>

Methods:
addItem(item: Item, quantity: int): boolean (Calls item.hasSufficientStock() before adding)
removeItem(itemId: int): void
calculateTotal(): double (Sums the totals of all CartItems)

2.4 David - Search Items + Reserve Cart

Tasks Completed

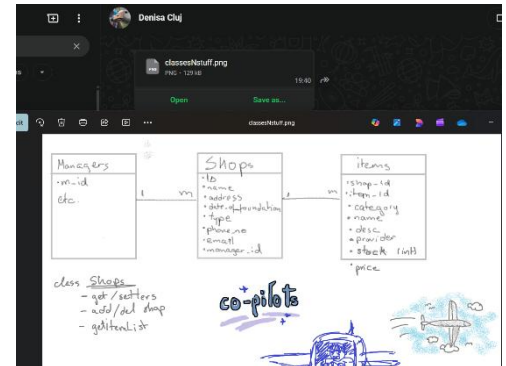
- Create database tables for given requirements
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- Create tasks for given requirements (Trello)



2.5 Denisa - Manage Shop Inventory

Tasks Completed

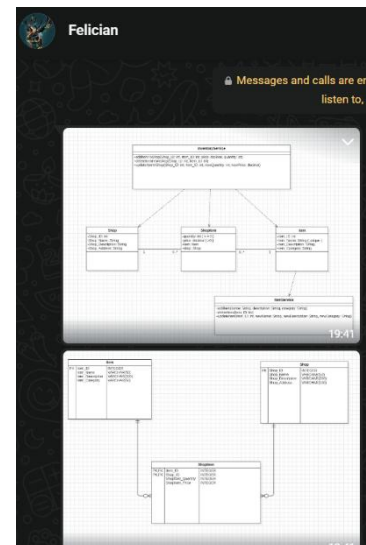
- Create database tables for given requirements
- Created classes for given requirements
- Create tasks for given requirements (Trello)



2.6 Felician - Manage Global Items

Tasks Completed

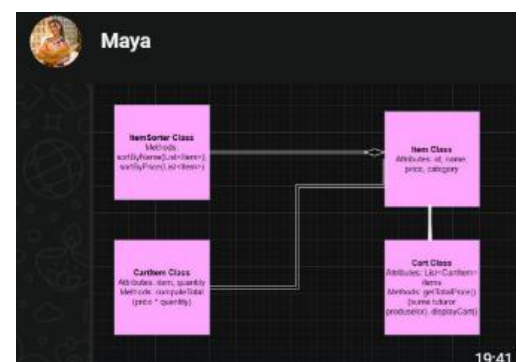
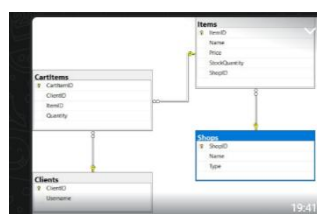
- Create database tables for given requirements
- Created classes for given requirements
- Create tasks for given requirements (Trello)



2.7 Maya - Sort Items + View Cart

Tasks Completed

- Create database tables for given requirements
- Created classes for given requirements
- Create tasks for given requirements (Trello)



3. Team Contributions

Tasks Completed

- Create Requirements list
- Create Use Case Diagram
- Create Database Diagram
- Create Class Diagram

Method of completion

- 3 online meetings where we worked on shared .drawio files(for the diagrams) and .dox file (for the requirements)

